Sections: B,C & D

Linear Algebra (MT-1004)

Homework#39

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Question 1

Part 1: Singular Value Decomposition (SVD)

Given the matrix

$$A = \begin{bmatrix} 4 & 0 \\ 3 & -5 \\ 1 & 2 \end{bmatrix}$$

(a) **Find the Singular Value Decomposition (SVD) of A:** Compute the matrices U, Σ, and V^T such that $A = U\Sigma V^T$.

(b) **Express A as a sum of rank-1 matrices:** Write A as a sum of rank-1 matrices of the form $\sigma_i u_i v_i^T$, where σ_i, u_i, and v_i are the singular values and singular vectors from the SVD.

(c) **Find the bases of the four fundamental subspaces of A:**

- Column space (Range of A)
- Null space (Kernel of A)
- Row space (Range of A^T)
- Left null space (Kernel of A^T)

Part 2: Matrix Norms

(a) **Compute the following norms:**

- $\|A\|_F$ (Frobenius norm)
- $\|A\|_2$ (Spectral norm)
- $\|A\|_*(NuclearNorm)$

(b) Explain the significance of each norm in applications such as data compression and error minimization.

Part 3: Left and Right Inverse

(a) **Determine if A has a left or right inverse:**

- If it does, compute the left inverse $A_L^\dagger$ or the right inverse $A_R^\dagger$ using SVD.